Flight Automation

Cal Poly Pomona | College of Engineering | Aerospace, Electrical and Computer, Mechanical and Chemical Engineering Departments

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Introduction

Unmanned Aerial Vehicles (UAVs) are becoming increasingly prevalent in , rapid deliveries and recognition of hard-to-reach locations. However, complications arise from initial flight sequences and ground-based landing stations that are often limited to specific drones, leading to miscommunication and reduced efficiency. The development of our GCS and BVM is a microcosm of attempts to provide drones with a physical "pit-stop" for recharging. Additionally, specialized controllers restrict access to autonomous features. The lack of flexibility observed in aircraft landing and controlling procedures contributes to decreased efficiency, battery life, miscommunication, and failure to operate.

Project Goals

Problem Statement: Communication limitations within drones can lead to ineffectiveness and failures in operations, hindering their ability to carry out tasks effectively and safely.

Constraints: Flying drones without the right preparation or design of system integration makes it difficult to control properly.

Solution: The purpose of our Flight Automation goals is to build, program, and research methods on constructing drones and solving flight patterns to implement these drones to autonomously land by procedure on the Ground Control Station and integrate with the Battery Vending Machine.

Development of Standard Quadcopter & DJI Tellos

Standard Quadcopter:
- Designed **flight architecture** for constructing drones [Figure 1a]
- Utilizes Pixhawk 4 flight controller with an open-source **autopilot system**
- Advanced autopilot capabilities for **autonomous flight**
- Displays **telemetry information** and has **drone configuration** settings
- Quadcopter model shown in [Figure 1b]

DJI Tellos:
- Leading commercial drone for **autonomous flight education**
- Equipped with a **camera** to capture aerial footage
- Our team developed a method to angle the camera downwards [Figure 1c] for better ground-level footage
- Uses **PID algorithm** with **OpenCV** to enhance autonomous flight capabilities

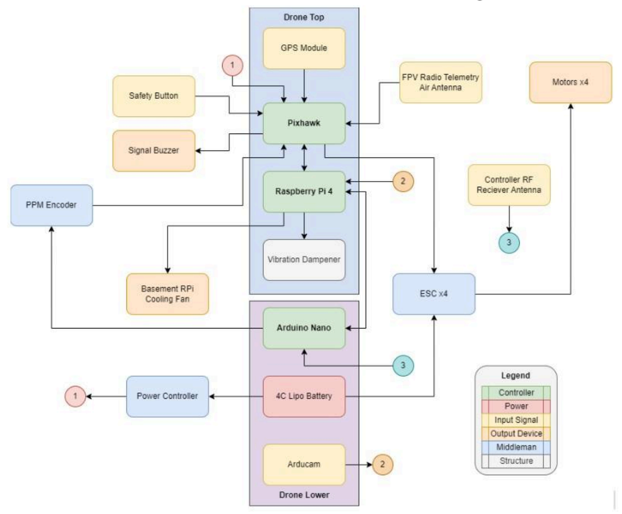


Figure 1a: Quadcopter Flight Architecture

Figure 1b: Standard Quadcopter

Figure 1c: DJI Tello Body and Top Down Camera Mounts

Hardware Development

We created a control system to operate our drones with manual and autonomous flight control mechanisms. This includes a reprogrammable controller and reconfigurable receiver designed to replicate the functionality of commercial equipment.
- The receiver and transmitter use **microcontrollers** based on the ATMEGA32U4 chip and the Arduino Pro Micro breakout board [Figure 2a].
- The ATMEGA32U4 chip has a 16MHz clock, 32kB of RAM, and 18 I/O pins for signal transmission.
- **Custom** receiver is built using a **PCB** breakout board for the Arduino Pro Micro, providing it with communication capabilities through an NRF24L01+PA+LNA wireless module and an OLED display.
- Commercial **transmitter** controllers operate on the 2.4GHz band and contain joysticks, potentiometer knobs, and switches for flight options.

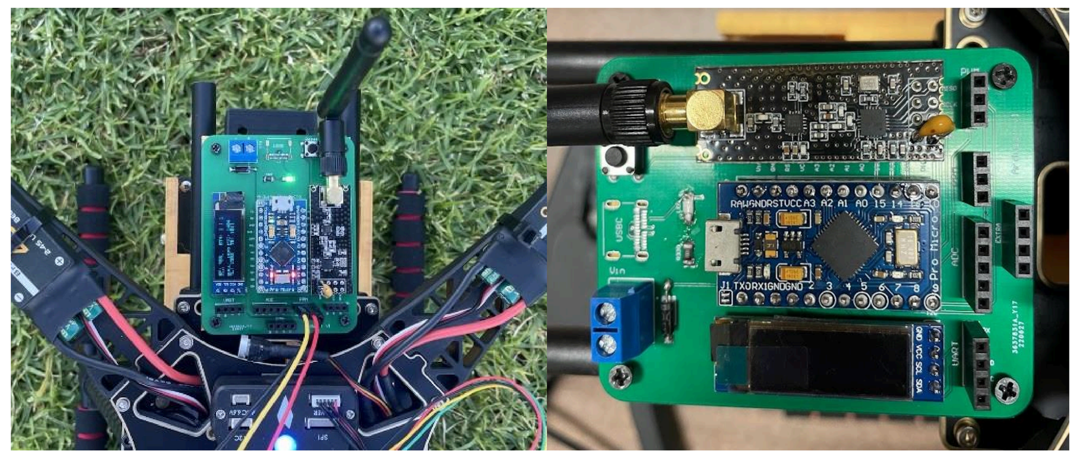


Figure 2: Custom Receiver on PCB Breakout board

Wireless Synchronization Method

UAV Synchronization Procedure
- NRF2401L transceiver module used for **wireless communication**
- GCS calculates **computational data** for **controller input**
- Reprogrammable **transmitters** and reconfigurable **receiver** allow **simultaneous control** of two flight architectures

The Resolution-Increasing Algorithm
- **Algorithm** used to increase sensitivity when controlling a drone
- Bit splitting, masking, and bitwise operations to **encrypt** and **decrypt data**
- Increases **control resolution** while retaining a small delay of 5ms per successfully transmitted packet
- Requires both ends of the communication to be **synchronized** on the same packet architecture

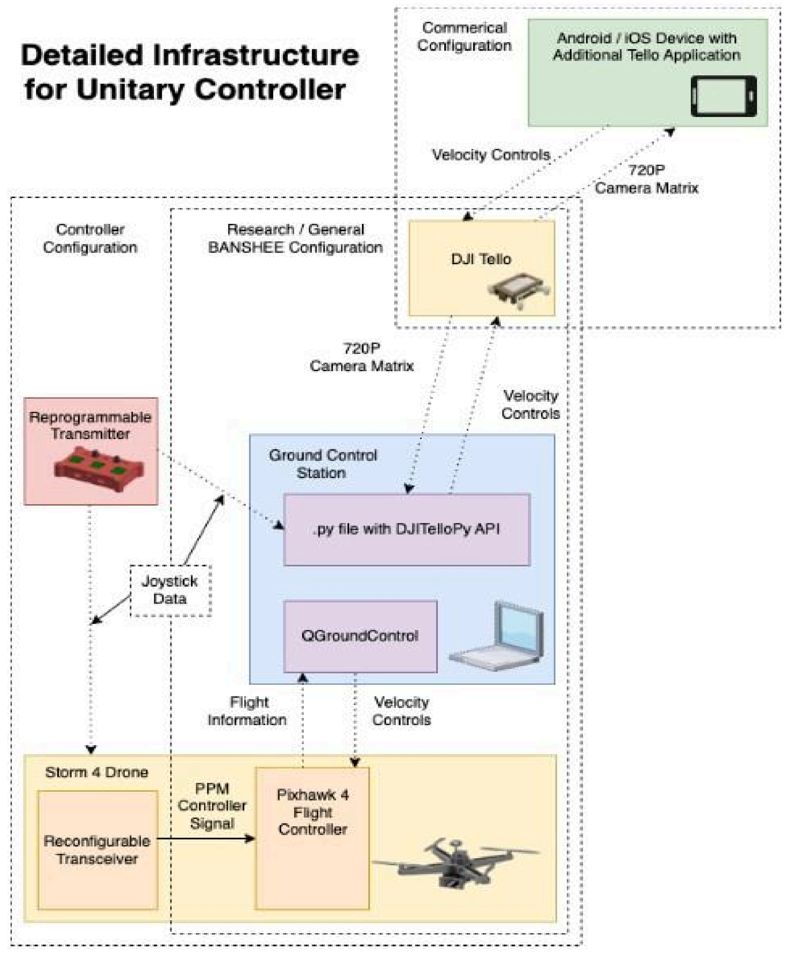
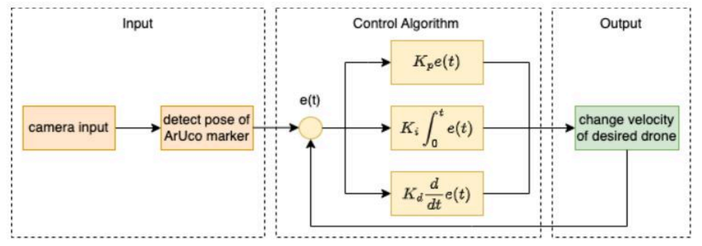
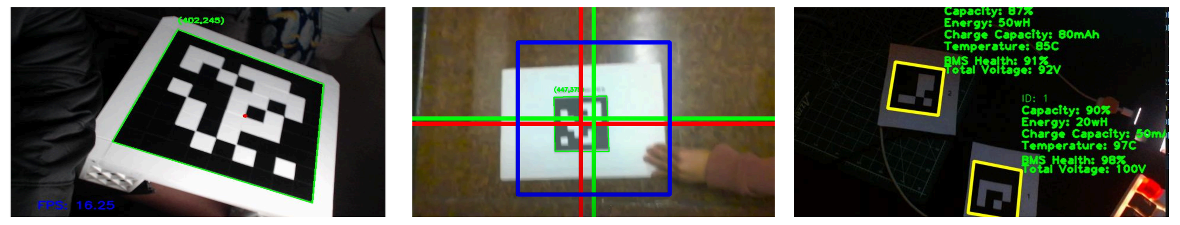


Figure 3: Design Infrastructure for Synchronized Controller

Autonomous Fiducial Marker Recognition

We implemented autonomous fiducial marker recognition using **AR marker detection, adaptive threshold,** and **grayscale methods.**
- ARTag marker system used for robust **marker detection**
- Configured for different operation, such as **drone landing**
- **Adaptive thresholding** to analyze historical data and monitor drone performance
- Grayscale simplifies algorithm and reduces computational requirements
- Methods enable multiple drones to read same marker simultaneously
- Synchronizes drones' behaviors, improving **efficiency** and **reliability**





Autonomous Landing Procedure

Our landing procedure was created within this flowchart [Figure 5b]
- **"Takeoff" mode** is initiated when the drone is at a distance greater than 12 inches from the ground charging station.
- In the absence of a signal from the ground station, the drone enters **"Fly" mode**, and follows commands for movement.
- If battery life becomes 50% or less and the ground station marker is not detected, the drone enters **"Search" mode.**
- **"Landing" mode** is activated when the drone detects the ground station marker.
- **OpenCV** and **image detection** are used to facilitate the landing process.
- The DJI Tello drone monitors **battery life, camera status,** and scans for **ArUco markers** to execute movement commands [Figure 5a].
- Fig 5c, The ground control station is a large **AR marker** that aids in **autonomous landing.**

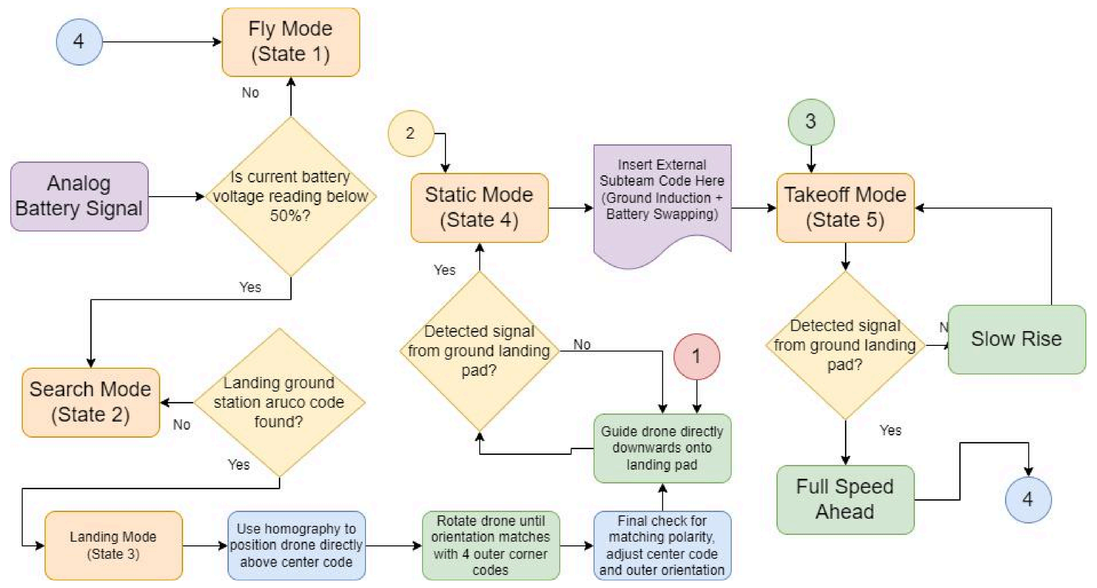


Figure 5a: Landing & Takeoff Flowchart

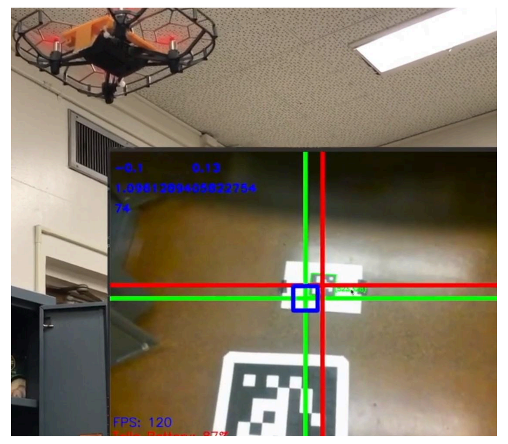


Figure 5b: DJI Tello Landing Test & Interface

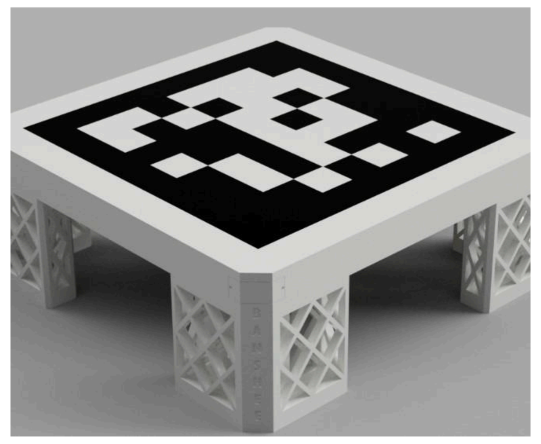


Figure 5c: ArUco Marker Ground Control Station

Test Flight Procedure

Ensuring the safety and readiness of our flight system is our top priority. To achieve this, we created flight checklist [Fig.] covering critical hardware calibration, structural verification, and controller validation. Our calibration process checks components such as the accelerometer, ESCs, and radio controller for proper functionality. Structural verification involves double-checking propeller tightness and safety switch activation. Finally, controller validation ensures that our flight controller is connected and ready for takeoff.

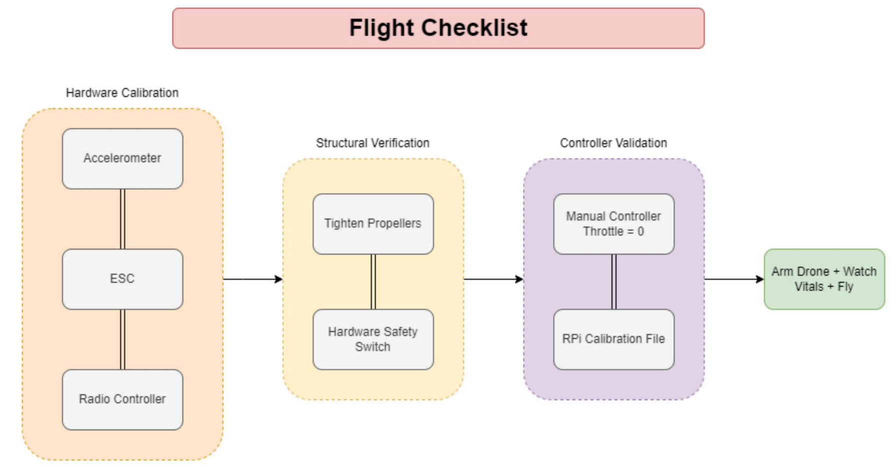


Figure 6: Pre Flight Checklist

Results

- Conducted two endurance tests with our quadcopter to evaluate power consumption impact of wireless synchronization technology
- Results showed that technology did not significantly reduce drone's power consumption
- Demonstrated a feasible platform for utilizing a single controller for multiple drone applications

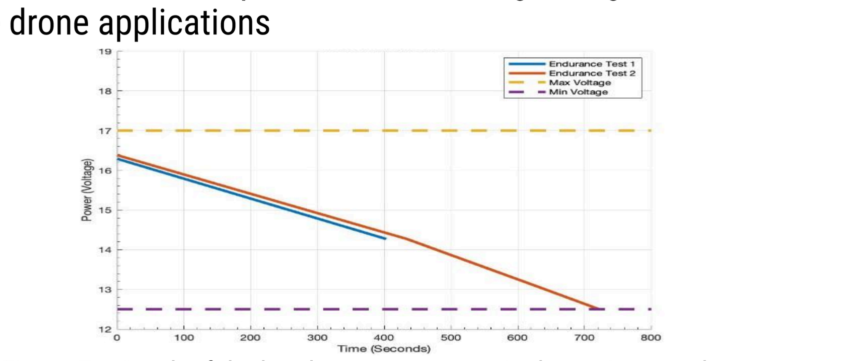


Figure 7a: Graph of depleted power over time in determining endurance

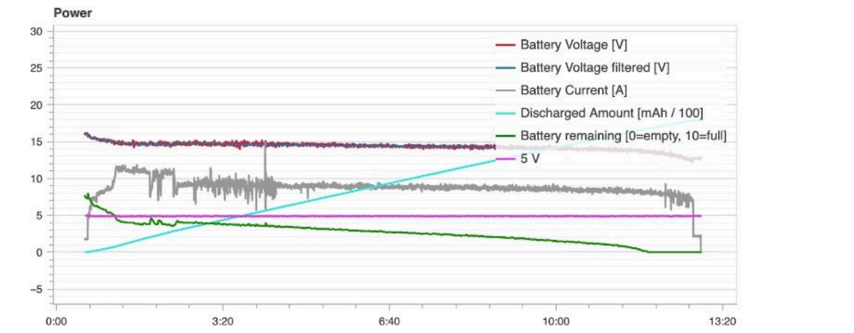


Figure 7b: Flight logger data from Pixhawk for Endurance testing

Conclusion

Our reprogrammable controller offers a quick and efficient solution for drone customization, as demonstrated by successful flight tests on two different UAVs. This standardized approach to takeoff and landing protocols, combined with auto-landing fiducial marker system integration, could enhance the efficiency and success of unmanned aerial vehicle operations in a range of contexts.

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